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└──────────┘ BRANCH 2: GAT graph (cross-sectional, RAW feats @ t)
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| node_raw = price[:, :, -1, :]           # LAST timestep RAW features, NOT
h_lstm |

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|           [B, N, 5]
|
|   -> GATLayer gat1 (5 -> 64, heads=4; concat)   [B, N, 256]
|
|   -> GATLayer gat2 (256 -> 64, heads=4; concat) [B, N, 256] = h_gnn
|
|   attention masked by (adjacency>0), softmax over SOURCE nodes,
LeakyReLU + ELU
|   graph OFF => adjacency := identity (self-loops only)
|

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|----- BRANCH 3: NEWS (per node) + per-ticker GATE
|
| news * news_mask   [B,N,seq,146]
|
|   -> Linear(146 -> 64) + ReLU
|
|   -> LSTM(64 -> 64, layers=2, dropout=0.2) -> last step -> news_hidden
[B,N,64]
|   -> gate = sigmoid(gate_logits[ticker_ids])      # scalar per ticker,
in (0,1)
|   -> gated_news = gate * news_hidden   [B, N, 64]
|

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          h_lstm[64]          h_gnn[256]          gated_news[64]
          |-----|-----|
          |
          v
      HEAD: concat -> h   [B, N, 384]
            Linear(384 -> 64) + ReLU + Dropout(0.2)
            Linear(64 -> 1)           -> raw   [B, N]
            v
      POSITIVITY FLOOR (denormalized space, eps=1e-6)
      denorm = raw*std + mean          (per-ticker
StandardScaler)
      floored = eps*softplus(denorm/eps) + eps (>= 0; target
is Parkinson VARIANCE)
      out     = (floored - mean) / std          [B, N]
(normalized space)

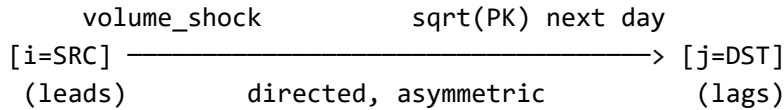
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Notes: - Branch 2 consumes RAW node features at the last timestep, not the LSTM hidden state. This keeps the graph branch an independent cross-sectional view (fair leave-one-out graph ablation) and matches the vol->PK edge semantics (volume_shock_i(t) -> sqrt(PK_j)(t+1)). - Without news (use_news=False), gated_news is zeros [B,N,64]; the gate is a no-op.

2. Directed vol->PK Top-5 edge

Adjacency built by lead-lag correlation, directed and asymmetric, frozen on TRAIN only.

For each TARGET ticker j , connect the Top-5 SOURCE tickers i ranked by:
 $\text{corr}(\text{volume_shock}_i(t), \sqrt{\text{PK}_j}(t+1))$ # source volume LEADS target volatility



$\text{adjacency}[i, j] = 1$ for i in Top5(j); diagonal self-loops kept
 $(\text{adjacency}[j, j] = 1)$

Attention (per GAT layer): mask entries where $\text{adjacency} \leq 0$, softmax over SOURCE j , aggregate.

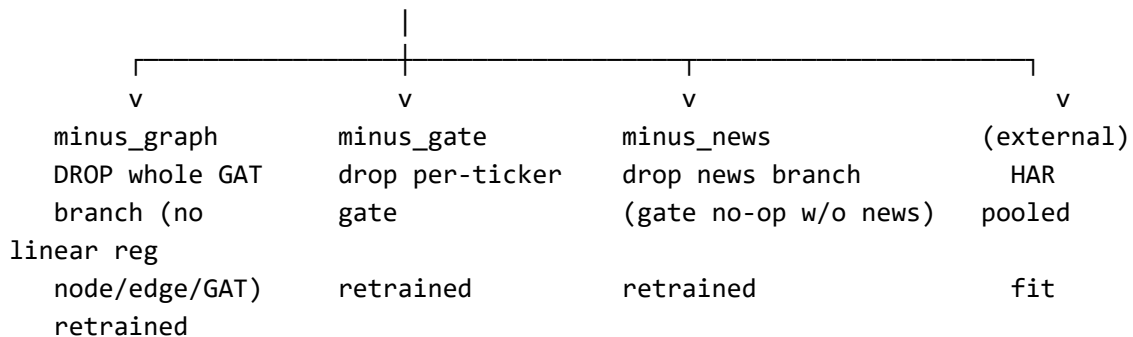
Chosen over undirected k-NN correlation-on-PK because: - (a) lead-lag is predictive/causal, not contemporaneous - (b) volume leads volatility - (c) directed captures who-leads-whom - (d) EDA-selected Top-5 (not an arbitrary k)

Leakage safety: edges are computed on TRAIN data only and frozen for val/test.

3. Leave-one-out ablation (run_ablation.py, h in {1, 5, 10, 22})

Build FULL, then RETRAIN each variant from scratch with exactly ONE component removed, so every effect is measured on the same footing (each variant trains in the same graph on/off regime it is evaluated in — no train/eval mismatch).

FULL (LSTM + vol->PK GAT graph + news + per-ticker gate)



Variant	LSTM	Graph (vol->PK)	News	Gate	Trained
HAR	-	-	-	-	pooled linear (ext)
FULL	yes	yes	yes	yes	retrained
minus_graph	yes	NONE (no GAT)	yes	yes	retrained
minus_gate	yes	yes	yes	OFF	retrained

Variant	LSTM	Graph (vol->PK)	News	Gate	Trained
minus_news	yes	yes	OFF	n/a	retrained

Contribution of component X (test QLIKE):

$\text{effect}(X) = \text{QLIKE}(\text{FULL}) - \text{QLIKE}(\text{minus}_X)$

negative => removing X hurt (QLIKE rose) => X HELPED

positive => removing X improved QLIKE => X did not help

Significance: Diebold-Mariano (HLN correction) on seed-ensembled test predictions, for FULL vs each minus_X and FULL vs HAR.

4. The 5 node features (shared by the LSTM and GAT branches)

Column order: [pk_daily, har_weekly, har_monthly, market_pk, volume_zscore(22)].

#	Feature	Definition	Causal/leakage
1	pk_daily	parkinson_volatility (the variance), clipped $\pm 3\sigma$ (train stats)	uses day t only
2	har_weekly	rolling-5 mean of pk	trailing
3	har_monthly	rolling-22 mean of pk	trailing
4	market_pk	cross-sectional MEDIAN of $\sqrt{\text{PK}}$ across present tickers at t (market factor)	contemporaneous (col t)
5	volume_zscore	$(\log_{10}(\text{volume}) - \text{roll.mean}) / \text{roll.std}$, rolling-22 window; no-volume tickers $\rightarrow 0.0$	trailing

Feature 5 uses a **22-trading-day (1-month) window**, unified with har_monthly's rolling(22) (TrackA-local override in run_trackA.build_trackA_basis; the underlying column key retains the legacy name volume_zscore_20 from the 2026-08-11 EDA baseline, which stays at window 20 so its recorded results remain reproducible). SEQ (LSTM lookahead) = 22. The GAT node input is the last-timestep feature vector, which already encodes multi-scale rolling context via features 2/3/5. ``